# Integrating Gradient and Mask-based Approaches for Vision Transformer Explainability

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***Abstract*—Vision Transformers (ViTs) have demonstrated outstanding performance across different computer vision tasks thanks to their self-attention mechanism that captures long-range dependencies effectively. However, the inherent complexity of ViTs presents significant challenges in explaining their outputs, which is fundamental in safety-critical domains. To tackle the challenge of explaining ViT outputs, this paper presents GradMask, a novel method that integrates gradients into the mask generation process to create explanation heatmaps. GradMask uses the query, key, and value matrices from each attention layer and computes their gradients with respect to a target class. Afterward, it uses these gradients to generate binary masks, which are then weighted by the corresponding ViT's confidence scores. Finally, it combines the weighted masks to generate the resulting heatmap. Experimental evaluations on an ImageNet subset with ViT and DeiT (Data-efficient Image Transformer) architectures show that GradMask achieves competitive performance according to standard explainability metrics, such as Insertion, Deletion, and Pointing Game. A hyperparameter analysis confirms the high computational efficiency of GradMask, while an ablation study highlights the importance of combining gradients and masks for the generation of the explanation heatmap. Finally, a qualitative analysis shows the improved explainability of GradMask compared to existing methods, making it a promising approach for understanding ViTs.**

***Index Terms*—Visual Explainability; Vision Transformer; Gradient-based Explainability; Self-Attention Mechanism; Computer Vision**

## I. Introduction

Vision Transformers (ViTs) [10] have demonstrated superior performance compared to traditional Convolutional Neural Networks (CNNs) across various computer vision tasks [13], [14]. By using the self-attention mechanism, ViTs effectively capture long-range dependencies, which makes them highly effective in tasks such as image classification, object detection, and segmentation. However, the complexity of ViTs introduces significant challenges in explaining their outputs, which is critical for several domains such as healthcare, autonomous vehicles, and safety-critical systems [17]. While some methods are available for explaining traditional deep learning models, the explanation of ViTs is still particularly difficult due to their reliance on self-attention mechanisms [3], [5]. The literature reports three categories of approaches for the explanation of ViTs, namely: *(i)* gradient-based, *(ii)* attention-based, and *(iii)* perturbation-based. Gradient-based approaches leverage the backpropagation of gradients to identify the contributions of input features to the model's predictions. Attention-based approaches work with the self-attention mechanisms of ViTs to track the information flow across Transformer layers. Finally, perturbation-based approaches identify important input regions by altering or masking specific parts of the input and analyzing the resulting changes in model predictions.

In this paper, we introduce GradMask, a novel explainability approach for ViTs, which integrates gradients from the self-attention components into the perturbation process to create explanation heatmaps, hence merging the contributions from the three research lines. By extracting query, key, and value matrices from each attention layer and calculating their gradients with respect to the target class, GradMask generates binary masks that identify the most important tokens. It then weights these masks using the ViT confidence scores and merges them to generate the final heatmap for the explainability of the ViT output. Following similar approaches in the literature [11], [25], we evaluated GradMask on a subset of ImageNet using ViT and DeiT architectures. Experimental results demonstrated that GradMask achieves promising performance compared to existing methods, as confirmed by the values of some standard explainability metrics, like Insertion, Deletion, and Pointing Game. The hyperparameter analysis further confirmed that GradMask maintains computational efficiency while achieving strong explainability. Additionally, an ablation study highlighted the role of the mask generation process in supporting GradMask's effectiveness. Finally, a qualitative analysis conducted on three examples of heatmaps generated by GradMask and several other approaches showcases the very good explanation capability of our approach.

This paper is structured as follows: Section II provides an

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overview of related work. Section III describes the proposed approach, while Section IV reports the experiments carried out to validate it. Finally, Section V draws our conclusions and outlines some possible future works.

## II. Related Work

Visual explainability methods for ViTs are commonly grouped into three categories, i.e., gradient-based, attention-based, and perturbation-based methods [22].

Gradient-based methods leverage the backpropagation of gradients to identify the contributions of input features or tokens to the model's predictions. Initially developed for CNNs, methods like Grad-CAM [20], Integrated Gradients [23], and SmoothGrad [21] have been adapted for ViTs. When applied to ViTs, these methods frequently incorporate data from attention matrices to exploit the self-attention mechanism inherent to Transformers. For example, [8] generates class-specific explanations by adjusting the contributions of attention heads based on gradients of the attention matrices, while also using residual connections and Layer-wise Relevance Propagation (LRP) [2] to determine the relevance of individual tokens. Similarly, Grad-Sam [4] integrates attention scores with gradients to assess the impact of particular inputs on the model's predictions. [7] introduces a framework that leverages gradients to determine the relevance of attention heads; it propagates information through the network using custom propagation rules designed to optimize both efficiency and contextual understanding. Additionally, Transition Attention Maps (TAM) [26] integrates gradients with a Markov chain model to examine information flow across ViT layers, allowing a more effective computation of token relevance. Furthermore, [9] presents a method that employs partial derivatives of the loss function with respect to tokens, splitting the explanation process into two phases, i.e., Attention Perception and Reasoning Feedback. Finally, [6] proposes a layer-wise approach that independently computes gradients for attention maps at each layer, enhances the signal through ReLU clipping, and combines these gradients across layers to produce relevancy maps highlighting the most significant regions of the input image.

Attention-based methods leverage the self-attention mechanisms of ViTs and aim to track the information flow across Transformer layers. For instance, [1] proposes attention-flow and rollout strategies to show how information is propagated within ViT models. Similarly, studies like [12], [16] focus on analyzing self-attention patterns to uncover the interactions between tokens influencing the model's decisions.

Perturbation-based methods take a different approach by assessing the significance of input regions through systematic alterations or masking of specific parts of the input, followed by an analysis of the resulting changes in model predictions. For instance, Randomized Input Sampling for Explanations (RISE) [18] identifies important regions by applying random masks to input segments and evaluating their impact. ViT-CX [25] focuses on combining embeddings from the final layer to generate saliency maps that emphasize the contributions of various input features. Finally, TIS [11] strategically introduces perturbations after the linear projection and positional encoding steps. In this way, it ensures dependable and artifact-free explanations while taking advantage of ViTs' ability to manage variable token counts.

## III. Proposed Approach

The objective of GradMask is to create a saliency map starting from the gradients of the main components of the self-attention mechanism, i.e., query, key, and value vectors. In self-attention, the query $Q$ represents the current token and determines how much attention it should give to other tokens. The key $K$ represents all tokens and measures their relevance to the query, essentially specifying how important each token is to the one making the query. The value $V$ carries the actual information of each token and is weighted by the relevance computed from $Q$ and $K$.

Let $T$ be a Vision Transformer model, $I$ an input image consisting of $B \times H$ pixels and belonging to the target class $c$. Let $T(I, c)$ be the confidence score of $T$ in predicting $I$ as belonging to $c$. When $I$ is given as input to $T$, it is divided into $N$ non-overlapping squared patches of size $s \times s$, making up a patch set $P$. Therefore, being $b = \frac{B}{s}$ and $h = \frac{H}{s}$, $I$ consists of $b \times h$ squared patches.

$T$ is composed of a set $L$ of $\lambda$ attention layers, and each attention layer $l \in L$ has its own query matrix $Q_l \in \mathbb{R}^{(N+1)\times E}$, key matrix $K_l \in \mathbb{R}^{(N+1)\times E}$, and value matrix $V_l \in \mathbb{R}^{(N+1)\times E}$, where $E$ is the embedding dimension. For each $l \in L$, we extract the corresponding matrices $Q_l$, $K_l$, and $V_l$, and remove the entry corresponding to the classification (CLS) token as the focus is exclusively on the patch contributions. Then, we compute the gradients of $Q_l$, $K_l$, and $V_l$ according to $T(I, c)$ and obtain three new matrices $\nabla Q_l \in \mathbb{R}^{N\times E}$, $\nabla K_l \in \mathbb{R}^{N\times E}$, and $\nabla V_l \in \mathbb{R}^{N\times E}$.

To effectively represent the information contained in an embedding matrix, we draw inspiration from the multi-head attention mechanism, where each head focuses on a specific portion of the data. Applying this idea to our scenario, we split the embedding dimension $E$ into $\epsilon$ partitions, each with a size $e = \frac{E}{\epsilon}$. This way of proceeding allows us to create masks on partitions of the embeddings, preventing situations where a few dominant features totally obscure the others. Therefore, each gradient matrix is split into $\epsilon$ matrices along the embedding dimension. These matrices are $\nabla Q_{l_i} \in \mathbb{R}^{N\times e}$, $\nabla K_{l_i} \in \mathbb{R}^{N\times e}$, and $\nabla V_{l_i} \in \mathbb{R}^{N\times e}$, $1 \leq i \leq \epsilon$. At this point, $\nabla Q_l$, $\nabla K_l$, and $\nabla V_l$ can be represented as:

$$\nabla Q_l = \nabla Q_{l_1} \oplus \nabla Q_{l_2} \oplus \cdots \oplus \nabla Q_{l_\epsilon} \tag{III.1}$$

$$\nabla K_l = \nabla K_{l_1} \oplus \nabla K_{l_2} \oplus \cdots \oplus \nabla K_{l_\epsilon} \tag{III.2}$$

$$\nabla V_l = \nabla V_{l_1} \oplus \nabla V_{l_2} \oplus \cdots \oplus \nabla V_{l_\epsilon} \tag{III.3}$$

where $\oplus$ is the operator that horizontally concatenates matrices having the same number of rows.

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From this point onward, the source of the gradient matrix becomes irrelevant, as we will treat all matrices in the same manner. Thus, we introduce a set $A_l$ of $3 \cdot \epsilon$ gradient matrices, comprising all the $\nabla Q_{l_i}$, $\nabla K_{l_i}$, and $\nabla V_{l_i}$ matrices. Each of these matrices has $N$ rows and $e$ columns. Let $\alpha_{l_j} \in A_l$ one of these matrices. Our approach generates a vector $a_{l_j}$ of $N$ elements corresponding to $\alpha_{l_j}$. The $n$-th element of this vector, $1 \leq n \leq N$, is obtained by summing the $e$ elements of the $n$-th row of $\alpha_{l_j}$. Formally speaking, we have that:

$$a_{l_j}[n] = \sum_{k=1}^{e} \alpha_{l_j}[n,k] \tag{III.4}$$

where $\alpha_{l_j} \in A_l$, $1 \leq j \leq 3 \cdot \epsilon$. Proceeding in this way, we obtain a set of vectors $a_{l_j} \in \mathbb{R}^N$ having a gradient value for each input patch of the image.

We can now generate binary masks by finding the most influential patches. To this end, given a vector $a_{l_j} \in A_l$, we define a threshold $\mu_{l_j}$ and set it to the mean of the $N$ elements of $a_{l_j}$. Formally speaking:

$$\mu_{l_j} = \frac{1}{N} \sum_{n=1}^{N} a_{l_j}[n] \tag{III.5}$$

Drawing inspiration from other approaches [11], we then create a binary mask $BM_{l_j} \in \mathbb{R}^N$ corresponding to $a_{l_j}$ as follows:

$$BM_{l_j}[n] = \begin{cases} 1, & \text{if } a_{l_j}[n] \geq \mu_{a_{l_j}} \\ 0, & \text{if } a_{l_j}[n] < \mu_{a_{l_j}} \end{cases} \tag{III.6}$$

In this way, we obtain $3 \cdot \epsilon$ binary masks, one for each gradient matrix $\alpha_{l_j} \in A_l$. These binary masks are then employed to create obscured versions of the input image. Each mask preserves only the tokens identified as significant through gradient analysis. This results in a set of modified images where different regions are emphasized, depending on the applied mask.

At this point, GradMask passes the obscured image $I'_{l_j}$, obtained by applying the binary mask $BM_{l_j}$ to $I$, through the ViT model to compute a confidence score $T(I'_{l_j}, c)$. This score specifies the confidence that $I'_{l_j}$ belongs to the target class $c$ and is fundamental to quantify the model's reliance on the image regions defined by each specific mask.

Afterward, GradMask weights each binary mask $BM_{l_j}$ by its corresponding confidence score and obtains a weighted mask $WM_{l_j}$ as follows:

$$WM_{l_j} = BM_{l_j} \cdot T(I'_{l_j}, c) \tag{III.7}$$

In order to generate the final explainability heatmap, GradMask repeats the previous process for all the layers. Then, it aggregates the weighted masks across all $L$ layers, all $\epsilon$ embedding partitions, and all the gradient components extracted from $Q$, $K$, and $V$. In this way, it obtains a vector $H_{I,c}$ as follows:

$$H_{I,c} = \sum_{l=1}^{L} \sum_{j=1}^{3 \cdot \epsilon} WM_{l_j} \tag{III.8}$$

Finally, since $H_{I,c} \in \mathbb{R}^N$, GradMask reshapes it to $\mathbb{R}^{b \times h}$, and applies bilinear interpolation to obtain a $B \times H$ image representing the resulting heatmap. This heatmap visually highlights the most influential regions in $I$ for class prediction, emphasizing the areas that most affect the classification of $I$ to the target class $c$.

The pseudocode algorithm describing the behavior of GradMask is reported in Algorithm 1.

**Algorithm 1** Pseudocode algorithm describing the behavior of GradMask

**Input**

- $T$: a ViT model
- $I$: an image to visually explain
- $c$: a target class
- $\epsilon$: the number of embedding partitions

**Output**

- $H_{I,c}$: the heatmap explaining $c$ in $I$

$WM = \emptyset$
Extract the set $L$ of $\lambda$ attention layers from $T$
**for** each layer $l \in L$ **do**
    Compute the matrices $Q_l$, $K_l$, and $V_l$ associated with the $l$-th layer of $T$
    Compute the gradient matrices $\nabla Q_l$, $\nabla K_l$, and $\nabla V_l$
    Split horizontally the gradient matrices into $\epsilon$ matrices $\nabla Q_{l_i}$, $\nabla K_{l_i}$, and $\nabla V_{l_i}$ and add them to $A_l$
    **for** each gradient matrix $\alpha_{l_j} \in A_l$ **do**
        Compute $a_{l_j}$ from $\alpha_{l_j}$
        $\mu_{l_j} = \frac{1}{N} \sum_{n=1}^{N} a_{l_j}[n]$
        Compute the Binary Mask $BM_{l_j}$ from $a_{l_j}$ by selecting the tokens having a value greater than $\mu_{l_j}$
        Create $I'_{l_j}$ by obscuring I using $BM_{l_j}$
        $WM_{l_j} = BM_{l_j} \cdot T(I'_{l_j}, c)$
        Add $WM_{l_j}$ to $WM$
$H_{I,c} = \sum_{l=1}^{L} \sum_{j=1}^{3 \cdot \epsilon} WM_{l_j}$
Reshape $H_{I,c}$ into a $b \times h$ matrix
Apply bilinear interpolation to $H_{I,c}$ to obtain a $B \times H$ matrix representing the final heatmap

## IV. Results and Discussion

### A. Experiment Setup

*1) Dataset:* Following prior studies [18], [1], [8], [7], [25], [11], we used a specific subset of the ImageNet validation dataset [19], comprising 5,000 images [25], [9].

*2) Models:* In our experiments, we employed Vision Transformer (ViT) [10] and Data-efficient Image Transformer (DeiT) [24] as Transformer models. The specific implementations are ViT-Base[1] and DeiT-Base[2].

*3) Evaluation Metrics:* We evaluated performance using the Insertion and Deletion metrics [18]. The Insertion metric measures how effectively the target class is identified as pixels are added in order of importance, with the "Area Under the Curve" (AUC) of the resulting curve measuring performance.

[1]https://huggingface.co/google/vit-base-patch16-224

[2]https://huggingface.co/facebook/deit-base-distilled-patch16-224

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The deletion metric measures the decrease in confidence as high-importance pixels are removed. The possible values of both metrics belong to the real range $[0, 1]$; the higher (resp., lower) the value of the Insertion (resp., Deletion) metric, the better the performance of the approach into evaluation. As noted by [11], these metrics taken individually do not represent an optimal measure. On the other hand, their difference is one of the most used metrics to evaluate the ability of the approach to identify the most significant regions. Following previous work [11], we applied four pixel obfuscation methods to calculate the Insertion and Deletion metrics, namely: *(i) Mean*, replacing pixel values with the image's mean value; *(ii) Blur*, applying pixel blurring; *(iii) Black*, replacing pixels with black; and *(iv) Rand*, substituting pixels with random colors. Additionally, we used the Pointing Game localization metric to measure the accuracy of identifying key image regions. Its values are in the real range $[0, 1]$ with higher values reflecting better localization.

*4) Compared Approaches:* The approaches compared against GradMask are TIS [11], ViT-CX [25], Chefer1 [8], Chefer2 [7], BT H [9], BT T [9], TAM [26], RISE [18], IntegratedGrad [23], and SmoothGrad [21]. We used the hyperparameter values suggested in their respective papers for these comparisons.

The interested reader can find the subset index list and the code implementing our approach at https://github.com/DavideTraini/GradMask.

### B. Quantitative Results

In this section's experiments, we set $\epsilon = 4$. In Table I, we report the Insertion, Deletion, and Insertion - Deletion metrics obtained by GradMask and the comparing approaches on the ViT-Base model. In this table, as in the others, we have bolded the best value and underlined the second-best value in each column. From the analysis of this table, we note that GradMask is the best performing approach according to the Insertion metric, where we observe an increase of 30.77% in Mean, 2.90% in Blur, 36.00% in Black, and 34.04% in Rand with respect to the second-best method. In the Deletion case, GradMask ranks second in Blur tied with TIS, while IntegratedGrad achieves the lowest Deletion values. However, as will be clear in Section IV-D, IntegratedGrad generates heatmaps very hard to explain. Finally, as for the overall metric Insertion - Deletion, GradMask is the best performing approach according to Mean, Black and Rand, while it is the second-best approach according to Blur. In particular, we note an increase of 19.05% in Mean, 25.00% in Black, and 18.42% in Rand with respect to the second-best approach.

Table II reports the Insertion, Deletion, and Insertion - Deletion results of GradMask and comparing approaches on DeiT-Base. From the analysis of this table, we observe that GradMask is the best approach according to Insertion and Insertion - Deletion. We note a higher Insertion of 29.82% in Mean, 8.82% in Blur, 33.33% in Black, and 40.74% in Rand with respect to the second-best approach. The values of the Deletion metrics obtained by GradMask are in line with most of the other comparing approaches except Rand. However, for the overall metric Insertion - Deletion, GradMask obtains the best values with an improvement of 9.52% in Mean, 10.00% in Black, 9.52% in Black, and 12.20% in Rand with respect to the second-best approach.

As the last experiment in this section, we computed the Pointing Game accuracy of GradMask and the comparing approaches using both ViT-Base and DeiT-Base. The corresponding results are reported in Table III. This table shows that GradMask achieves the highest Pointing Game results. We observe an increase of 2.92% in the ViT-Base and of 6.30% in the DeiT-Base with respect to the second-best approach. Taking into account the semantics of the Pointing Game metric specified above, this means that GradMask is the most accurate approach in highlighting the regions in an image that align with the target class. The results of this experiment are in line with the ones we observed in Tables I and II.

### C. Hyperparameter Analysis and Ablation Study

We conducted a hyperparameter analysis and an ablation study on GradMask. In this section, for the sake of space, we only report the results using ViT-Base as Transformer model; similar results can be obtained on DeiT-Base.

We tested GradMask using different values of $\epsilon$, corresponding to a different number of embedding partitions, hence a different number of generated masks. In Table IV, we report the Insertion, Deletion, and Insertion - Deletion results of GradMask with 4 values of $\epsilon$. From the analysis of this table, we observe that the higher $\epsilon$, the higher the GradMask performance.

However, the higher $\epsilon$, the higher the computational cost of GradMask. To evaluate this factor, we computed the GFLOPs (Giga FLoating point Operations) of GradMask with the same values of $\epsilon$. We also computed this metric for the comparing approaches. The corresponding results are reported in Table V. From the analysis of this table we observe that GradMask has a computational cost comparable to the one of the other approaches. Specifically, it ranges from 1,264.27 GFLOPs for $\epsilon = 1$ to 4,742.28 GFLOPs for $\epsilon = 4$, which is significantly lower than the cost of TIS and slightly lower than the one of VIT-CX. While some other methods exhibit minimal GFLOPs requirements, the performance improvements achieved by GradMask with respect to them justify the modest increase in computational load.

Finally, we performed an ablation study by removing the aggregation of the weighted masks for the generation of the resulting heatmap. Following similar approaches in the literature [6], we created a heatmap using the gradients of $Q$, $K$, and $V$ without the generation of the corresponding binary and weighted masks. Therefore, for each image patch, we summed and then normalized the gradients of $Q$, $K$, and $V$, reshaped the gradient matrix, and finally interpolated it to generate the heatmap. In this way, we compared GradMask with an approach based only on gradients (called NoMask). We computed the Insertion, Deletion, and Insertion - Deletion metric, and reported the corresponding results in Table VI.

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TABLE I: Results of the Insertion, Deletion, and Insertion - Deletion metrics of GradMask and the comparing approaches - ViT-Base model

| | *Insertion* ↑ | | | | *Deletion* ↓ | | | | *Insertion - Deletion* ↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Method* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* |
| GradMask | **0.68** | **0.71** | **0.68** | **0.63** | 0.19 | 0.39 | 0.19 | 0.18 | **0.50** | 0.32 | **0.50** | **0.45** |
| TIS [11] | 0.52 | 0.66 | 0.50 | 0.47 | 0.10 | 0.39 | 0.10 | 0.09 | 0.42 | 0.28 | 0.40 | 0.38 |
| ViT-CX [25] | 0.51 | 0.61 | 0.41 | 0.39 | 0.20 | 0.42 | 0.14 | 0.18 | 0.28 | 0.20 | 0.31 | 0.35 |
| TAM [26] | 0.43 | 0.61 | 0.41 | 0.39 | 0.14 | 0.43 | 0.14 | 0.13 | 0.28 | 0.18 | 0.27 | 0.26 |
| Chefer1 [8] | 0.42 | 0.61 | 0.41 | 0.39 | 0.15 | 0.44 | 0.14 | 0.13 | 0.28 | 0.17 | 0.27 | 0.26 |
| Chefer2 [7] | 0.43 | 0.61 | 0.41 | 0.39 | 0.15 | 0.44 | 0.14 | 0.13 | 0.28 | 0.17 | 0.27 | 0.26 |
| BT H [9] | 0.45 | 0.63 | 0.43 | 0.41 | 0.12 | 0.41 | 0.12 | 0.11 | 0.33 | 0.21 | 0.32 | 0.30 |
| BT T [9] | 0.46 | 0.62 | 0.44 | 0.42 | 0.13 | 0.42 | 0.12 | 0.11 | 0.33 | 0.21 | 0.32 | 0.30 |
| RISE [18] | 0.46 | 0.62 | 0.45 | 0.42 | 0.16 | 0.45 | 0.16 | 0.15 | 0.30 | 0.17 | 0.29 | 0.27 |
| IntegratedGrad [23] | 0.19 | 0.69 | 0.16 | 0.15 | **0.08** | **0.31** | **0.06** | **0.06** | 0.11 | **0.38** | 0.10 | 0.08 |
| SmoothGrad [21] | 0.37 | 0.59 | 0.36 | 0.35 | 0.10 | 0.45 | 0.10 | 0.09 | 0.27 | 0.14 | 0.26 | 0.26 |

TABLE II: Results of the Insertion, Deletion, and Insertion - Deletion metrics of GradMask and the comparing approaches - DeiT-Base model

| | *Insertion* ↑ | | | | *Deletion* ↓ | | | | *Insertion - Deletion* ↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Method* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* |
| GradMask | **0.74** | **0.74** | **0.76** | **0.76** | 0.28 | 0.41 | 0.27 | 0.30 | **0.46** | **0.33** | **0.46** | **0.46** |
| TIS [11] | 0.57 | 0.65 | 0.57 | 0.54 | 0.15 | 0.40 | 0.15 | 0.14 | 0.42 | 0.25 | 0.42 | 0.41 |
| ViT-CX [25] | 0.51 | 0.61 | 0.51 | 0.48 | 0.20 | 0.42 | 0.20 | 0.18 | 0.31 | 0.19 | 0.31 | 0.30 |
| TAM [26] | 0.50 | 0.59 | 0.50 | 0.46 | 0.23 | 0.45 | 0.23 | 0.19 | 0.27 | 0.14 | 0.26 | 0.26 |
| Chefer1 [8] | 0.51 | 0.60 | 0.51 | 0.48 | 0.22 | 0.45 | 0.22 | 0.18 | 0.29 | 0.15 | 0.29 | 0.29 |
| Chefer2 [7] | 0.50 | 0.60 | 0.50 | 0.47 | 0.23 | 0.45 | 0.23 | 0.19 | 0.28 | 0.14 | 0.27 | 0.28 |
| BT H [9] | 0.52 | 0.60 | 0.52 | 0.49 | 0.19 | 0.43 | 0.19 | 0.16 | 0.33 | 0.18 | 0.33 | 0.33 |
| BT T [9] | 0.52 | 0.60 | 0.51 | 0.48 | 0.19 | 0.43 | 0.19 | 0.16 | 0.33 | 0.17 | 0.32 | 0.32 |
| RISE [18] | 0.55 | 0.61 | 0.55 | 0.52 | 0.25 | 0.46 | 0.25 | 0.21 | 0.30 | 0.15 | 0.30 | 0.31 |
| IntegratedGrad [23] | 0.32 | 0.68 | 0.30 | 0.28 | **0.14** | **0.38** | **0.12** | **0.13** | 0.18 | 0.30 | 0.18 | 0.15 |
| SmoothGrad [21] | 0.45 | 0.62 | 0.43 | 0.43 | **0.14** | 0.44 | 0.14 | **0.13** | 0.31 | 0.18 | 0.30 | 0.31 |

TABLE III: Results of the Pointing Game metric of GradMask and the comparing approaches

| *Method* | *ViT-Base* | *DeiT-Base* |
|---|---|---|
| GradMask | **0.880** | **0.877** |
| TIS [11] | 0.823 | 0.825 |
| ViT-CX [25] | 0.700 | 0.700 |
| TAM [26] | 0.737 | 0.635 |
| Chefer1 [8] | 0.768 | 0.748 |
| Chefer2 [7] | 0.727 | 0.654 |
| BT H [9] | 0.855 | 0.775 |
| BT T [9] | 0.846 | 0.755 |
| RISE [18] | 0.753 | 0.766 |
| IntegratedGrad [23] | 0.633 | 0.297 |
| SmoothGrad [21] | 0.499 | 0.742 |

Analyzing this table, we observe that our process of mask generation increases the performances of GradMask. For instance, the improvement of the Insertion - Deletion metric achieved by GradMask with respect to NoMask ranges from 25.00% to 35.14%.

### *D. Qualitative Results*

This section provides a qualitative analysis of the heatmaps generated by GradMask, comparing them to the results of several related approaches. For this comparison, we used the ViT-Base model to generate heatmaps for three randomly selected ImageNet images from different classes. The resulting visualizations are shown in Figure 1. From the analysis of this figure, we observe that GradMask is able to identify the most relevant regions according to the target class. In the "spoonbill" row, we note that our approach accurately highlights the shapes of the spoonbill, while giving much less relevance to the background. In the "upright piano" row, GradMask assigns very high importance to the shape of the piano, while giving no attention to the player. Additionally, it generates the most focused heatmap, whereas other methods tend to emphasize the piano player and background, as indicated by the prominent red and yellow areas in their heatmaps. Finally, in "street sign" row, GradMask correctly highlights the two street signs present in the image, while it gives less relevance to the building; instead, other approaches generate less focused heatmaps that do not capture relevant features. Overall, the generated heatmaps seen in this section align with the results presented in the previous section, further confirming

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TABLE IV: Results of the Insertion, Deletion, and Insertion - Deletion metrics of GradMask using different values of $\epsilon$ - ViT-Base model

| | *Insertion* ↑ | | | | *Deletion* ↓ | | | | *Insertion - Deletion* ↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\epsilon$ | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* |
| 1 | 0.66 | 0.69 | 0.66 | 0.62 | 0.24 | 0.41 | 0.23 | 0.23 | 0.43 | 0.28 | 0.43 | 0.38 |
| 2 | **0.68** | 0.70 | **0.68** | **0.63** | 0.21 | **0.39** | 0.21 | 0.20 | 0.47 | 0.30 | 0.47 | 0.42 |
| 3 | **0.68** | 0.70 | **0.68** | **0.63** | 0.20 | **0.39** | 0.20 | 0.19 | 0.49 | 0.31 | 0.49 | 0.44 |
| 4 | **0.68** | **0.71** | **0.68** | **0.63** | **0.19** | **0.39** | **0.19** | **0.18** | **0.50** | **0.32** | **0.50** | **0.45** |

TABLE V: GFLOPs required by GradMask and the comparing approaches

| *Method* | *GFLOPs* |
|---|---|
| GradMask ($\epsilon = 1$) | 1,264.27 |
| GradMask ($\epsilon = 2$) | 2,423.61 |
| GradMask ($\epsilon = 3$) | 3,582.94 |
| GradMask ($\epsilon = 4$) | 4,742.28 |
| TIS [11] | 17,808.70 |
| RISE [18] | 140,927.36 |
| VIT-CX [25] | 4,918.71 |
| Chefer1 [8] | 377.67 |
| Chefer2 [7] | 105.13 |
| BT H [9] | 2,264.73 |
| BT T [9] | 2,262.70 |
| TAM [26] | 2,203.77 |
| IntegratedGrad [23] | 3,572.95 |
| SmoothGrad [21] | 3,572.93 |

TABLE VI: Results of the Insertion, Deletion, and Insertion - Deletion metrics of GradMask with and without masks - ViT-Base model

| | *Insertion* ↑ | | | | *Deletion* ↓ | | | | *Insertion - Deletion* ↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Method* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* | *Mean* | *Blur* | *Black* | *Rand* |
| NoMask | 0.60 | 0.67 | 0.60 | 0.58 | 0.23 | 0.42 | 0.23 | 0.22 | 0.37 | 0.25 | 0.37 | 0.36 |
| GradMask | **0.68** | **0.71** | **0.68** | **0.63** | **0.19** | **0.39** | **0.19** | **0.18** | **0.50** | **0.32** | **0.50** | **0.45** |

that GradMask achieves very satisfying performance.

### *E. Discussion*

After showing the experimental results, we describe some advantages and limitations of our approach.

As for the advantages, unlike other perturbation-based approaches (e.g., [11], [18], [25]), GradMask generates masks using the gradients of the query, key, and value matrices with respect to a target class. This ensures that both the binary and weighted masks are directly linked to the object of interest. In contrast, other methods rely only on the ViT's confidence score. The combination of gradient-based and perturbation-based techniques in GradMask enables the generation of heatmaps strongly tied to the target class, thanks to the gradient information infused into the initial binary masks. Another advantage of GradMask is its consideration of all self-attention components for gradient computation, including the value matrix $V$. In contrast, other approaches (e.g., [8], [7], [9]) focus only on the attention matrix, computed as $Q \cdot K^T$, while neglecting the contributions of the matrix $V$. By incorporating the matrix $V$, GradMask retains all the information processed in the self-attention mechanism. This comprehensive approach allows GradMask to reach high performance, as it leverages both the relational structure from $Q$ and $K$ and the contextual content encoded in $V$.

However, GradMask has also some limitations. For example, it currently cannot be applied to hierarchical ViTs, such as Swin [15]. In Swin-like architectures, each attention layer generates masks of different sizes, due to the hierarchical image processing. As a result, GradMask should generate weighted masks with different sizes, which is not currently possible. To address this challenge, we need to upscale the masks to a uniform size while retaining their information. Another limitation regards token selection during the creation of binary masks. In fact, when computing gradients with respect to a target class, gradient explosion may cause some values to become outliers. This can negatively impact the computation of $\mu$, resulting in incorrect token selection. To mitigate this issue, the function $\mu$ could be replaced with alternative functions more robust to outliers.

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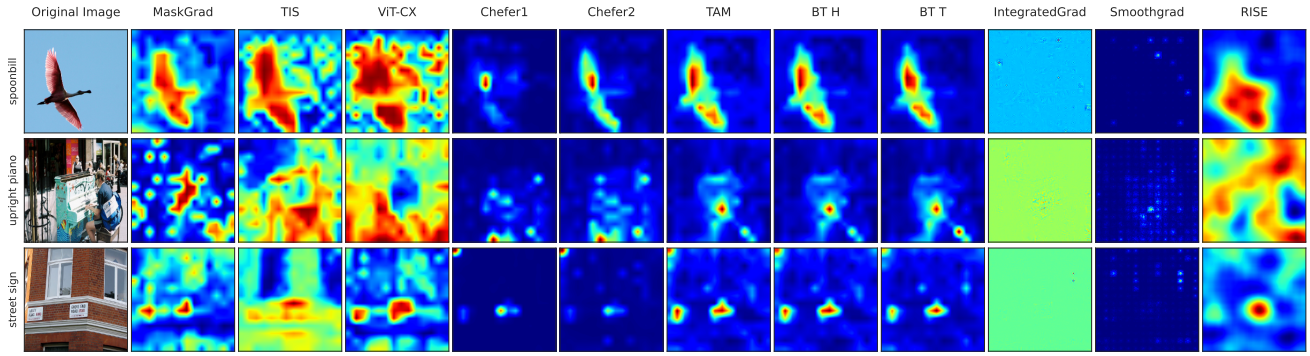



Fig. 1: Comparison of the heatmaps extracted by GradMask and related approaches for three random ImageNet images using ViT-Base

## V. Conclusion

In this paper, we have proposed GradMask, a novel explainability approach for Vision Transformers. The novelty of GradMask lies in its capability of integrating gradients within the perturbation process to construct an explanation heatmap. GradMask works by extracting the query, key, and value matrices from each attention layer of a Vision Transformer. It then computes the gradients of these matrices with respect to the target class and splits them along the embedding dimension. This process creates multiple masks representing subcomponents of the embeddings. Afterward, for each split gradient matrix, GradMask generates binary masks that highlight the tokens with the highest impact. Each binary mask corresponds to an obscured image, which is used to weight the corresponding regions based on the confidence score returned by the Vision Transformer when that mask is provided as input. In this way, GradMask obtains a set of weighted masks and then merges them to construct the final heatmap.

We tested GradMask on a subset of the ImageNet validation set and on two Vision Transformer architectures (ViT and DeiT), achieving very promising results. Specifically, we compared our approach with several related ones and found that it outperforms them, achieving the best performance according to the Insertion - Deletion and Pointing Game metrics. Additionally, we conducted a hyperparameter analysis that highlighted the impact of $\epsilon$ on explainability performance and computational cost. Finally, we highlighted the contribution of mask generation in GradMask through an ablation study and conducted a qualitative analysis.

Finally, we think that GradMask could be extended in several directions. For instance, we plan to extend its concepts and way of proceeding to the explainability of Natural Language Processing (NLP) Transformers. Moreover, we aim to adapt GradMask to multimodal Transformers that process multiple inputs, such as images and text. In such cases, it would be necessary to consider gradients from all input types, requiring an approach that integrates the different modalities effectively. Finally, we intend to explore interactions between the gradient matrices using network analysis. Indeed, the matrix multiplication $\nabla Q \cdot \nabla K^T$ returns the squared attention matrix, which can be represented as a graph. By modeling $\nabla V$ as a feature of this graph, we hope to exploit concepts and techniques of network analysis to further enhance the explainability of Vision Transformers.

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